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Strategic implications for Industrial OEMs  
transitioning to Product-as-a-Service business  
model

# Abstract:

Many OEMs in Industrial Equipment domain are moving from product selling model to outcome-based model, where equipment is leased out to customers based on either (a.) production-based selling – customers would be charged based on per unit output OR (b.) outsourcing model – equipment/machines are rented out to customers on per hour usage or on the Overall Equipment Effectiveness (OEE) model. This new business model is called Equipment-as-a-Service (EaaS) or Product-as-a-Service (PaaS) model, where OEMs are embracing Servitization where equipment is used as service-business model. Although, this concept is not new to the industry, for example, Rolls Royce introduced the engine leasing model a few decades ago, we have seen increase in adoption of PaaS business model in the past few months. This shift is mainly driven by three factors – consumer demand, advancements in IIoT technologies, and new revenue streams for OEMs. In this whitepaper, we will discuss the impact of PaaS business model on OEM's existing operations and understand how OEMs can reap the benefits through the adoption of the PaaS proposition.

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# Introduction

Product-as-a-Service, or PaaS, is a model of renting machinery to manufacturers on a pay-per-use or subscription basis. Unlike the Product Selling model, where OEMs only used to design, manufacture, and sell the equipment, PaaS enables OEMs to offer aftermarket services by entering into long-term contracts with end customers for maintenance, spare parts, annual servicing, upgradation, and recycling of products. The transition from one-time CAPEX sales to a recurring revenue model is shown below:

The market for PaaS (for discrete manufacturing industries) is growing with CAGR of 35% and expected to reach \$131B in 2025 from \$22B in 2019. Now, more companies expect that delivering services would be a more significant part of their business in the next 2 to 3 years. PaaS is now moving from a niche alternative to becoming an integral part of business strategy for OEMs.

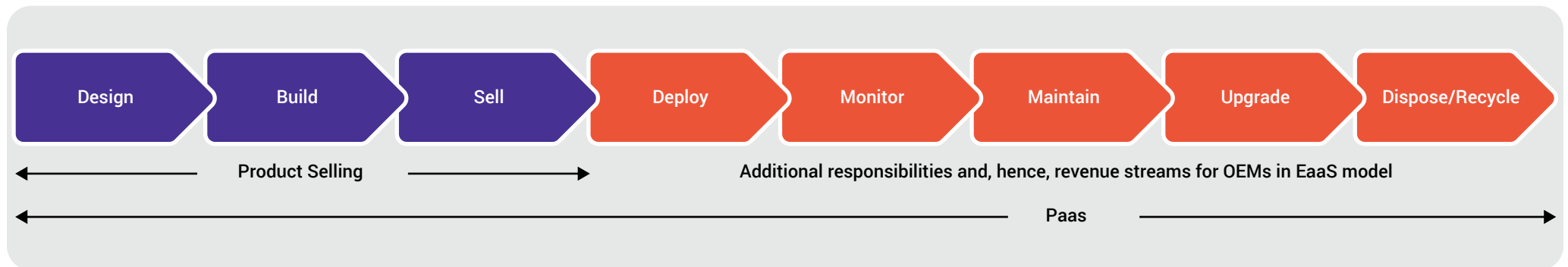


Figure 1: Product-as-a-Service model for OEMs



# PaaS is becoming a strategic imperative for OEMs

The transition to PaaS model is mainly driven by below factors:

## **Customer Demand (manufacturing organizations)**

Customers have started demanding PaaS services from OEMs for the reasons stated below:

- Shift from CAPEX to OPEX based business models – Onset of Covid has brought in restrictions on organization in terms of infrastructure expansion through CAPEX investments. Also, customers want to make their bottom lines and cash flow streams more predictable and resilient to crises. Hence, customers are looking for avenues to move to OPEX based business models.
- Flexibility – Customers are increasingly looking for flexible manufacturing setup (either owned or captive) where output can be easily modified based on consumer demands.

- Risk of ownership and Total Cost of Ownership reduction – customers now expect OEMs to make provisions for uptime & performance guarantees and take ownership of outcome responsibility by servicing long term maintenance contracts until the end of the product lifecycle.

## **Advancements in Industrial IoT and Digital Technologies**

Advancements in technologies such as IoT, Cloud, 5G, Big Data and AI, which were not very accessible ten years ago, have given rise to many digital solutions and services that are either improving adoption of EaaS models – or enabling them in the first place. A good example is the integration of an IoT layer (communication + sensor + cloud) on existing products. This enables sharing of product usage data to OEMs via a cloud-based platform through which customers can be charged based on actual usage or production output.

## **Aftermarket services revenue**

Driven by global competition, uncertain economic environment and pandemic conditions, most industry segments are experiencing decline in asset margins and topline in terms of annual sales volumes. This has prompted OEMs to tap into aftersales revenues. [Average operating margin from aftermarket business globally is 2.5 times the operating margin from new equipment sales.](#) Capturing aftermarket services revenue could help OEMs stabilize the revenue streams and improve profitability.

Below we highlight some examples that help understand how the PaaS model can unlock revenue synergies for both OEM and Customer alike.

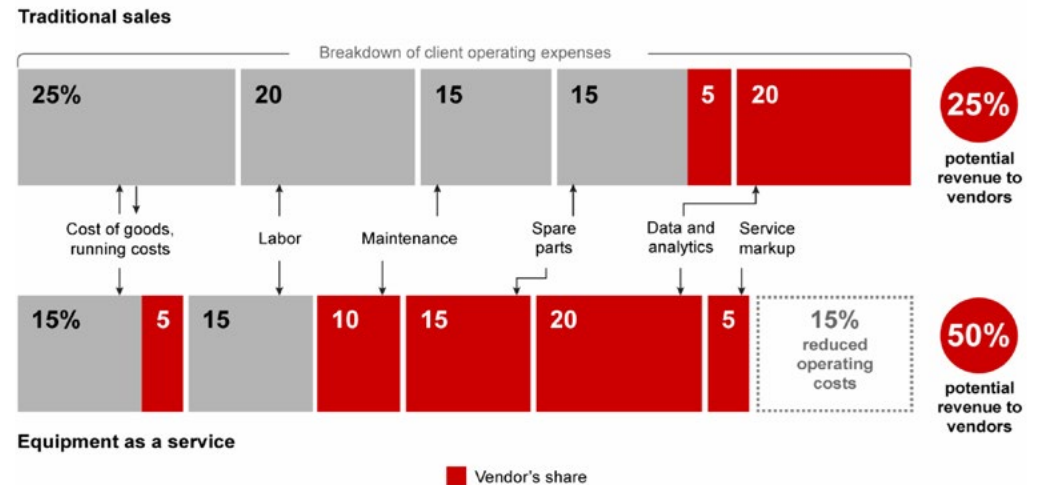


| OEM                                                                                                                                                                                                                                                                                                                                                                                | Sales Value | Financing and Usage based billing | After sales revenue (maintenance and spares) | Additional Services | Data Analytics and Digital Services | Total Revenue                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------|----------------------------------------------|---------------------|-------------------------------------|--------------------------------------------------------------------|
| <b>Product Selling approach for 3 years : Limited contact, No control over asset</b>                                                                                                                                                                                                                                                                                               |             |                                   |                                              |                     |                                     |                                                                    |
| OEM Revenue (\$)                                                                                                                                                                                                                                                                                                                                                                   | 150000      | -                                 | 18000                                        | 8000                | 6000                                | 182000                                                             |
| Profit Per contract (\$)                                                                                                                                                                                                                                                                                                                                                           | 12000       | -                                 | 5400                                         | 2400                | 2400                                | 22200                                                              |
| <b>Product as a Service (Product operating cost 3000 monthly) : 5 years Long Term contact, Complete Control Over Asset</b>                                                                                                                                                                                                                                                         |             |                                   |                                              |                     |                                     |                                                                    |
| OEM Revenue (\$)                                                                                                                                                                                                                                                                                                                                                                   | 3000        | 5000                              | 30000                                        | 12000               | 10000                               | 237000<br>(considered for 5 years)                                 |
| Profit Per contract (\$)                                                                                                                                                                                                                                                                                                                                                           |             | 1000                              | 15000                                        | 6000                | 5000                                | 27000<br>(considered for 5 years, 1000 financing charges per year) |
| Assumptions:<br>8 to 10 % profit on Product Selling<br>25 to 30% margin on after sales and services for product selling and 50%+ for PaaS<br>30 to 40% margin on data analytics and digital services, 50%+ for PaaS<br>Financing and Usage based billing – additional benefits of financing can be availed by OEMs as this model helps customers to reduce the liabilities and TCO |             |                                   |                                              |                     |                                     |                                                                    |

Table 1: Illustrative example for differentiation between Product Selling vs Product-as-a-Service model

In 1st approach, where OEMs are selling products with service contract for 3 years (assuming product life is 5 years), OEMs earn a revenue of \$182,000, with a profit of \$22,000. Customers have to service for initial CAPEX of 150,000 and look after equipment for remaining lifespan (2 years). In 2nd approach, OEMs manage the equipment until end of lifecycle. In this case, OEMs can earn more revenue (and margins) from aftermarket services. Customers shift from CAPEX to OPEX based model paying \$3000 per month and saving on capital infrastructure costs. Customers can effectively manage their cash flows and have flexibility in manufacturing process. Also, the risk of downtime and breakdown is mitigated by long term service contracts.

By moving to PaaS model, OEMs can earn more revenue and margin through aftermarket services and usage-based pricing. Service relationship can reduce the operating expenses for customers as most of the capital expenses of product and after services expenses will be taken care by the OEMs. Similar co-relation is given in [Bain & Company's report](#), where PaaS (EaaS in the report) can enable more revenue streams for OEMs (vendors in report shown in red color) by reducing overall operating costs for customers (shown in grey color).



Source: Bain &amp; Company

Figure 2: Revenue and Margin improvements for OEMs and Customers

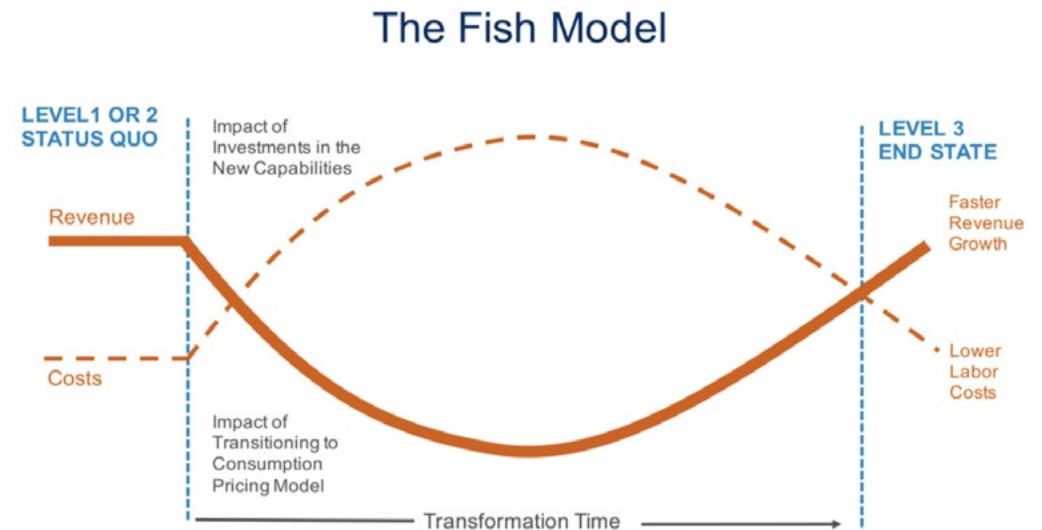


# Key requirements for OEM to shift to PaaS model

OEMs planning to transition to PaaS model need to thoroughly prepare the organization for sales and operations overhaul, provision for insurance and financing mechanisms and digital readiness. We must first understand the critical requirements for building a successful PaaS offering.

## Sales and Operations overhaul

- Transition from unit sales contract to continuous service contracts has profound implications on the financials, sales cycles, and customer service
- Sales incentives need to be aligned with service contracts and usage-based revenues
- Customer service will have to undergo big changes- shifting from only technical support post sales to becoming a close partner that helps manage the equipment, as well as ensures improved performance through diagnostics and data analysis.
- For efficient customer service, OEMs can forwardly integrate with distributors.
- The transition to PaaS model can put pressure on firm financials, especially topline, as revenues drop drastically in the early years. Moreover, OEMs will have to invest in product re-engineering costs to integrate IoT and Digital stack with products (sensors, software, communication protocols, data analytics etc.) to successfully deliver services model. This transition was experienced by technology companies when they moved to Software-as-a-Service models. The financial transition was termed as “[swallowing the fish](#)” as it is a function of rising costs and decreasing revenues.



Source: Technology-as-a-Service Playbook 2016 (TSIA)

Figure 3: Swallowing the Fish



### Financing and leasing models

- Estimating and managing cash flow cycles during the initial transition phase can be challenging for companies, as traditional banks may not be equipped with relevant services for PaaS models. Also, financial institutions may lose a steady and secure source of revenue from asset financing, as customers will no longer need loans for capital payments.
- Over the last few years, financial institutions, and banks with focus on PaaS business models have emerged. For example, MunichRe provides Special Purpose Vehicles (SPVs) to transfer assets from OEM's balance sheet, which is jointly managed by both parties.
- IFRS 16 and ASC 842 accounting regulations allow companies to move towards a PaaS model. For example, Siemens Financial Services supports customers with PaaS model.
- OEMs must include project and transformation financing beyond hardware stack. Additionally, they should retrofit the insurance that covers malfunctions and breakdowns during PaaS transition.

### Aftermarket levers

- OEMs need to calculate aftermarket lifetime value to trigger various improvement levers that focus on improving the core aftermarket revenues. For example, OEMs can increase product lifetime by re-marketing used equipment, modernizing existing equipment and re-pricing spare parts more dynamically.

### Digital Readiness

There are many industrial equipment manufacturers, for example - Hilti, Kaeser Compressors, Caterpillar etc., that have successfully introduced PaaS models, while numerous organizations have also failed in the transition. Successful organizations had very strong Software, IoT and Digital backbone. These organizations made provisions for below IoT and digital infrastructure to capture data from equipment that has critical implications on business decisions.

- Data Capture Mechanisms from existing legacy and new equipment IoT Devices - RFIDs, scanners, sensors
- Integration with enterprise data systems such as ERP and MES to lifecycle data such as orders, services, maintenance, payments, PoS



- Enabled smart products with communication infrastructure – wired and wireless communication technologies such as Bluetooth, NFC, satellite communication
- Modular and scalable cloud infrastructure as solution backbone
- Data management capabilities along with AI/ML powered analytics and predictive maintenance applications
- Use case enablement such as asset tracking, condition monitoring, digital twins, and automated usage-based billing for different user personas such as operator, plant manager, quality manager etc.

In a nutshell, the technology that connects machines and captures sensor data for service and maintenance insights has made Product-as-a-Service models a realistic option.

A schematic layout of digital backbone is shown below:

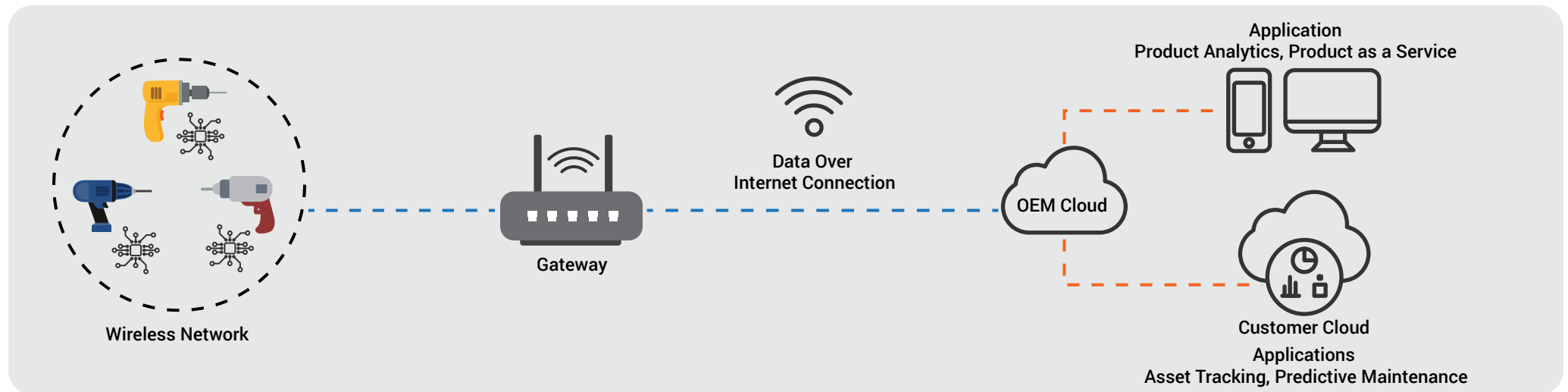


Figure 4: Digital infrastructure enablement



# Case Studies and Examples

Below are examples of few of the OEMs, which moved to PaaS offering.

| OEM                                                          | Situation                                                                                                                                                                                                                                     | Solution                                                                                                                                                                                                             | Benefits                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hilti</b><br><b>Power Tool-as-a-Service</b>               | <ul style="list-style-type: none"> <li>Low margins because of competition from China</li> <li>Rising demand for customized solutions</li> <li>Customers hit by financial crisis</li> </ul>                                                    | <ul style="list-style-type: none"> <li>Launched fixed monthly fee fleet management solution.</li> <li>Focus shifted from selling drill machines to selling holes</li> </ul>                                          | <ul style="list-style-type: none"> <li>Helped increase Hilti's sales by 26% and operating profit by 12%</li> <li>Leased more than 1 million tools to 100,000 different customers</li> <li>Cash flow optimization for the customer</li> </ul>                                  |
| <b>Caterpillar</b><br><b>Equipment-as-a-Service</b>          | <ul style="list-style-type: none"> <li>Not enough after-sale revenue because of disappointing service contracts</li> <li>Supply of cheaper aftermarket non-OEM parts accounted for 30-50% of spare parts affecting overall revenue</li> </ul> | <ul style="list-style-type: none"> <li>More customer outcome focused business model</li> <li>Customized contracts with customers</li> <li>Telematics IoT to monitor performance and maintenance</li> </ul>           | <ul style="list-style-type: none"> <li>Developed long term contracts with customers</li> <li>Improved profitability through aftermarket service revenues</li> <li>Premium pricing on OEM spare parts, with 25% to 30% of equipment revenue coming from parts sales</li> </ul> |
| <b>Sandvik Coromant</b><br><b>Machine Tools-as-a-Service</b> | <ul style="list-style-type: none"> <li>Covid-19 disrupted demand for equipment</li> <li>Customers changed their models from Capex to Opex</li> <li>Product-based selling with less focus on services</li> </ul>                               | <ul style="list-style-type: none"> <li>Digitization of products – connectivity and cloud platform</li> <li>Bundled selling contracts – Products + Services</li> <li>Consolidation of distribution network</li> </ul> | <ul style="list-style-type: none"> <li>Improved product designs from equipment data</li> <li>Long term contracts with customers</li> <li>Services based business model</li> <li>More focus on direct selling</li> </ul>                                                       |



## Benefits for OEMs

Though implementation of PaaS model comes with risks initially, the model brings multifold benefits for OEM organizations in terms of:

- Revenue maximization: OEMs can capture a large share of aftermarket revenue, which was initially left for third party services
- Profitability and cashflow improvement: aftermarket services have higher margins compared to products with stable and long-term cash flow
- Collaborative relationships with customers: OEMs must work closely with customers to improve product efficiency that will help them gain long term contracts with customers
- Control over equipment: Connected products enable control over the production parameters and OEMs can prevent the failures and breakdowns in advance. The data from products can be utilized to optimize next generation product designs.
- Network optimization: OEMs can integrate with regional hubs for spare parts, distribution, installation, and setup partners and keep the operational costs to a minimum level.
- Pricing and terms management: As the OEM carries more risks (transportation, equipment breakdown), they could gain more control over pricing strategy and command premium for value added services.

## Conclusion

PaaS models can fundamentally change the way industrial OEMs do business. Organizations from different equipment segments are trying to gain first mover advantage. They are likely to capture outsized rewards from the shift by developing closer and long-term relationships with their customers. To tap this blue ocean opportunity, OEMs should understand customers' PaaS-specific ambitions, evaluate their key priorities, and strategize the relevant action plan. As finance and digital infrastructures play key roles in the transition, OEMs should develop an ecosystem of Software & IIoT providers and financial service providers.



## About the Author

Amol has a decade of experience in the Manufacturing and Automotive industries. At Sasken, he works for the Product Engineering Practice and is responsible for developing and marketing new services and offerings for multiple domains.

## About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to-market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation.





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## Strategic implications for Industrial OEMs transitioning to Product-as-a-Service business model

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